



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

CANDIDATE NAME

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CIVICS GROUP

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H1 BIOLOGY

Paper 2 Structured and Free-response Questions

8876/02

25 August 2021

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
Section A	
1	/ 17
2	/ 18
3	/ 10
Section B	/ 15
Total	/ 60

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the synthesis of polypeptides on the rough endoplasmic reticulum.

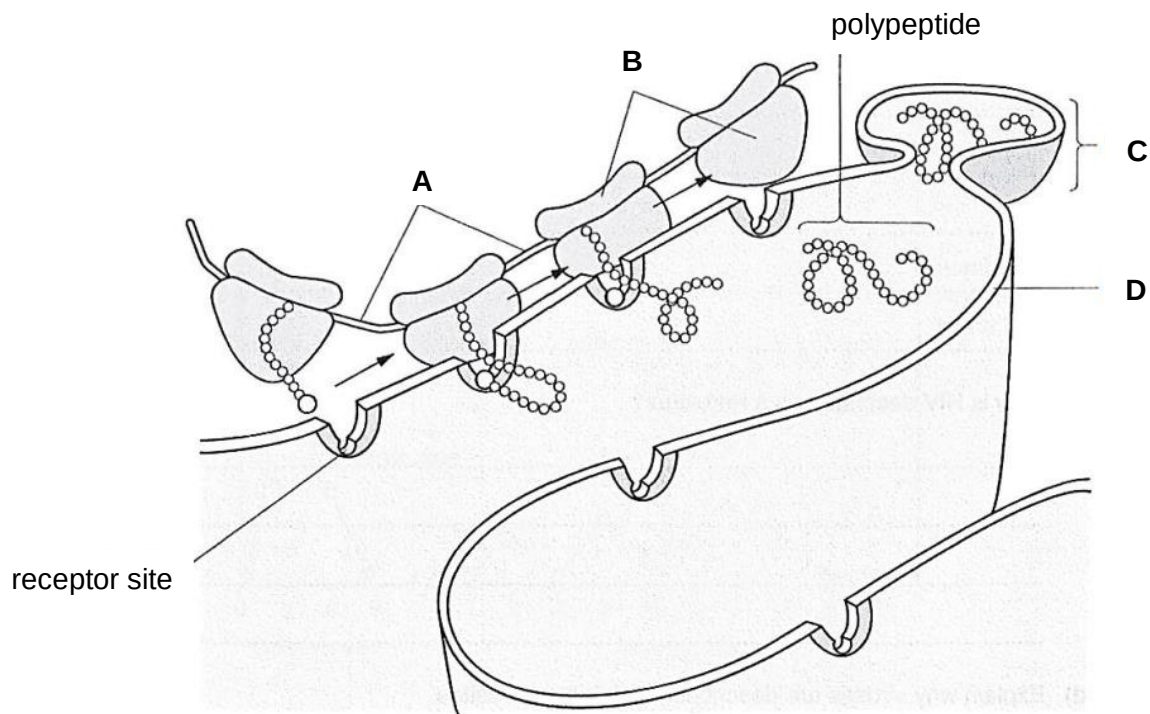


Fig. 1.1

- (a) Identify structures **A**, **B**, **C** and **D**, as shown on Fig. 1.1.

A

B

C

D

[4]

- (b) Use Fig. 1.1 to suggest how the structures labelled **B** attach to the endoplasmic reticulum through the growing polypeptide.

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..... [2]

An example of a polypeptide chain synthesised into the lumen of the endoplasmic reticulum is that of the hexosaminidase polypeptide chain. This will eventually give rise to the hexosaminidase enzyme, which is an enzyme found in lysosomes (lysosomal enzyme) that digests sphingolipids, a type of lipid.

- (c)** Describe what happens to the hexosaminidase polypeptide chain when it is released from the rough endoplasmic reticulum to eventually give rise to the hexosaminidase enzyme found in lysosomes.

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..... [3]

- (d)** Tay-Sachs disease is a genetic disorder that results in the destruction of nerve cells in the brain and spinal cord due to an accumulation of sphingolipids. It is normally caused by a genetic mutation in the hexosaminidase gene, which codes for the hexosaminidase enzyme.

One common mutation found on the hexosaminidase gene is the G444R mutation, where a glycine amino acid is replaced by an arginine amino acid at the 444th codon.

- (i)** State the type of mutation that has occurred.

.....[1]

The G444R mutation occurs at a splice site at the end of exon 11.

- (ii)** Explain how the G444R mutation can result in Tay-Sachs disease.

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.....[4]

A proposed treatment for Tay-Sachs disease is enzyme replacement therapy, where a solution containing the functional hexosaminidase enzyme is injected into the bloodstream of the patient. However, this treatment method is not very effective, as the enzyme is unable to be taken into the lysosome.

(iii) Suggest why the hexosaminidase enzyme is unable to be taken into the lysosome.

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..... [1]

Tay-Sachs disease is a recessive condition that is commonly found in the Ashkenazi Jews population. Studies have found that this population is subjected to a stronger selection pressure by *Mycobacterium tuberculosis* (the bacterium that causes tuberculosis) due to crowded environments, and carriers of the mutated hexosaminidase allele have increased resistance towards *Mycobacterium tuberculosis*.

(e) With reference to the information provided, explain how the mutated hexosaminidase allele is preserved in the population.

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..... [2]

[Total: 17]



- 2** Modern horses are a group of mammals adapted to living in the grassland. Their evolution can be reconstructed from the fossil record.

Table 2.1 shows the evolution of the teeth and height of the modern horse over 50 million years.

Table 2.1

Name	Teeth	Height / cm	Environmental conditions		
<i>Equus</i> (2 mya – today)	“Herbivorous teeth”	150	Predator present.	Grassland region. Mostly fibrous grass.	Edible insects extinct.
<i>Meryohippus</i> (26 mya)	Long cubical molars and flat grinding surfaces	100	Predator present.	Evolution of grass. Forest started to recede.	Edible insects becoming extinct.
<i>Mesohippus</i> (35 mya)	“Omnivorous teeth”	60	Predator present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground and upper part of the shrubs.
<i>Eohippus</i> (50 mya)	Short, bumpy molars	25	Predator present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground.

mya - million years ago

- (a)** With reference to Table 2.1, explain how natural selection led to the evolution of the horse over time.

..... [5]

With increasing global temperature due to rising carbon dioxide concentration in the atmosphere, one scientist hypothesises that grasses will benefit more than forest vegetation as they are better adapted to the temperature increase. A possible reason could be that the photosynthetic enzymes of grasses had an optimum temperature that is higher than the current average mid-day temperature of 30°C in the tropical grassland regions.

To test his hypothesis, the scientist carried out a study to study the effect of temperature on the growth of the grasses. Different patches of grass were subjected to different temperature inside experimental chambers. For all the experiments, other variables such as the species of grass, light intensity and nutrient solution were standardised.

At the end of the experiment, the changes in mass of the grass were calculated.

Table 2.2

Treatment	Mass of grass at the start of experiment / mg	Mass of grass at the end of experiment / mg	Change in mass as a percentage of the mass of grass at the start of experiment / %
30°C (Control)	1.24×10^4	1.53×10^4	
36°C	1.43×10^4	1.87×10^4	
42°C	1.44×10^4	1.74×10^4	

(b) (i) Calculate the percentage change in the mass of grass and complete the last column of Table 2.2. [1]

(ii) In the space provided below, show all your working for the percentage change in the mass of grass at 42°C.

[1]

(iii) State the optimum temperature of the enzymes.

.....[1]

(iv) Discuss whether the results obtained by the scientist supported his hypothesis.

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..... [4]

The cell wall of plant cells is made up of mainly cellulose. Cellulose synthase is responsible for the synthesis of cellulose.

Fig. 2.3 shows the cross-section of the enzyme embedded within the cell surface membrane of plant cells.

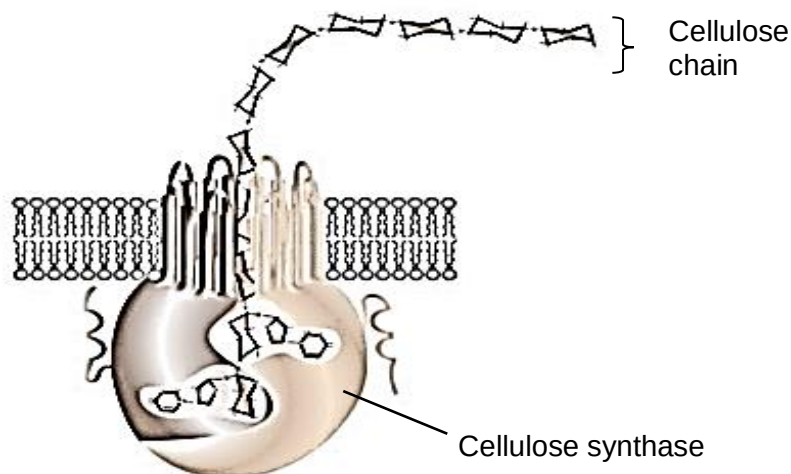


Fig. 2.3

(c) (i) In the space below, show how three monomers of cellulose are joined with each other to form a part of the cellulose chain.

Your answer should include the name of the bond that joins two adjacent monomers.

Cellulose synthase is part of a complex. Studies have shown that each complex is made up of six clusters, while each cluster consists of several cellulose synthase enzymes as shown in Fig. 2.4.

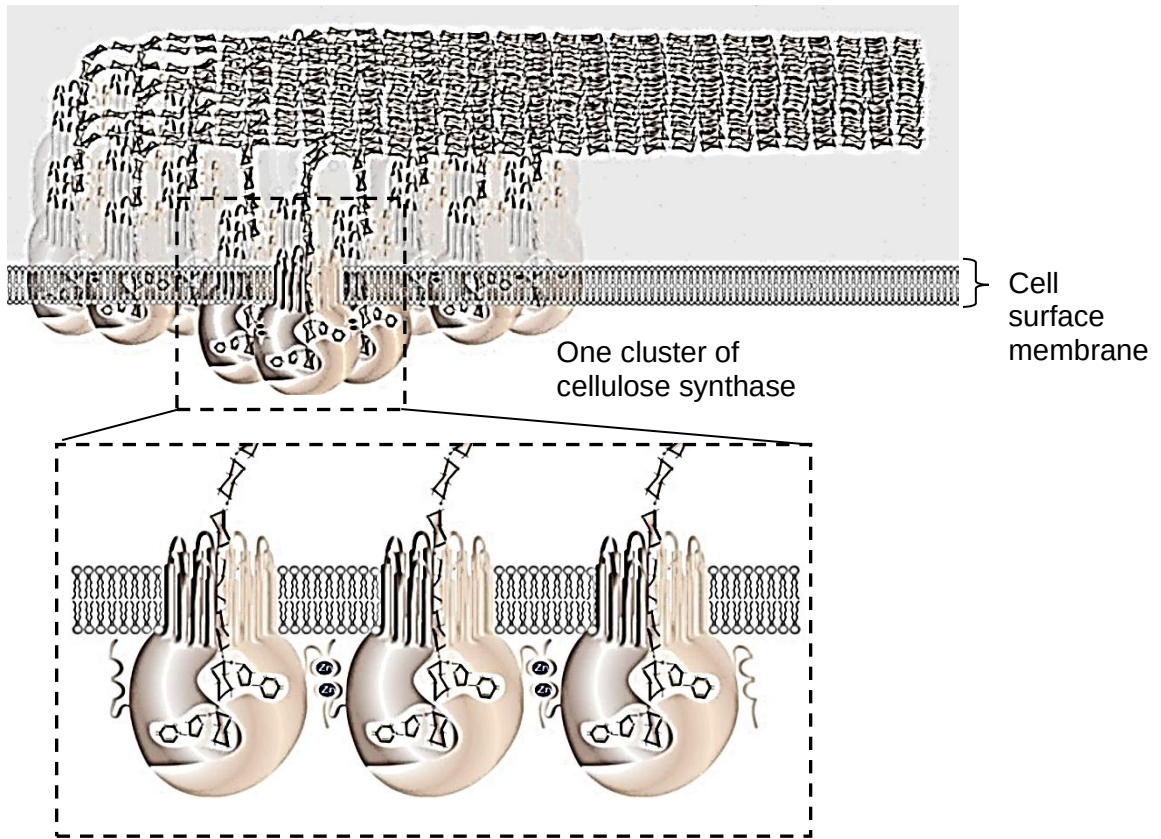


Fig. 2.4

- (ii) With reference to Fig. 2.4 and the role of cellulose, explain why cellulose synthase enzymes tend to be clustered in close proximity with each other.

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..... [3]

[Total: 18]

- 3 Antigens are found on human red blood cells and are used to determine the human ABO blood group system. The type and presence of antigens are determined by three alleles, I^A , I^B and I^O . Alleles I^A and I^B are co-dominant and both alleles are dominant to allele I^O .

(a) Define the term co-dominance using the example of the ABO blood group.

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..... [2]

There is another type of blood group system known as the Rhesus blood group, which is also genetically controlled. The gene for the Rhesus blood group has two alleles. The allele for Rhesus positive, Rh^+ , is dominant to that for Rhesus negative, Rh^- . Fig. 3.1 shows the inheritance of the Rhesus blood group in one family.

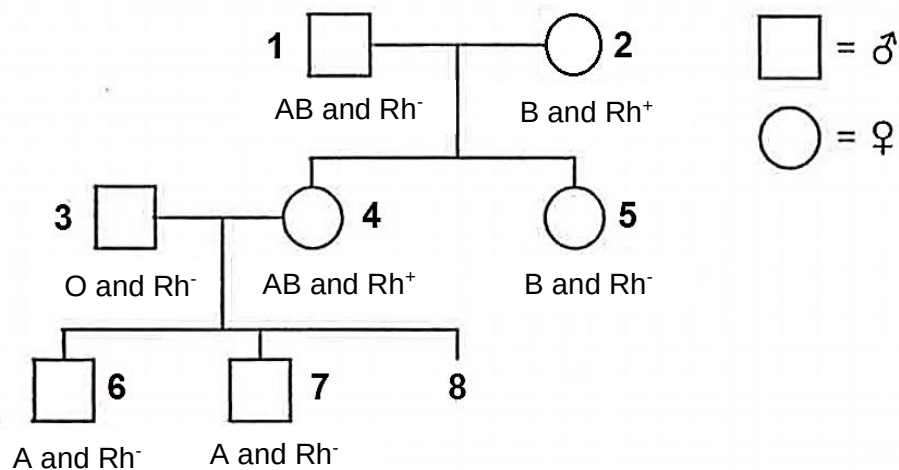


Fig. 3.1

- (b) (i) Explain one piece of evidence from Fig. 3.1 which shows that the gene for Rhesus blood group is **not** on the X chromosome.

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..... [2]

- (ii) Calculate the probability that individual 8 will be a boy with blood group B and Rh positive.

[2]

Fig. 3.2 shows the movement of chromatids in some blood cells that have just started metaphase of mitosis.

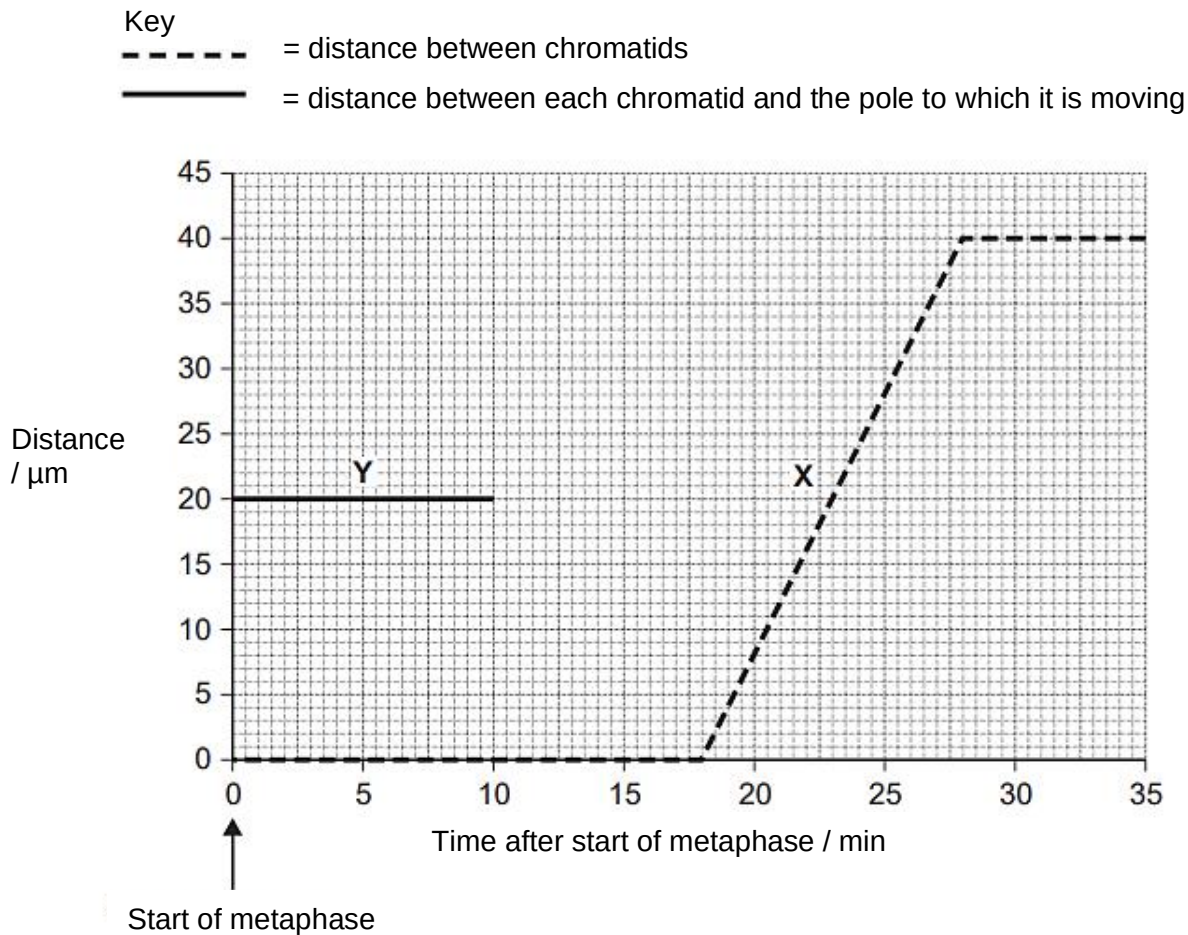


Fig. 3.2

(c) Complete line **Y** on Fig. 3.2 to show the expected distance between each chromatid and the pole to which it is moving. [2]

(d) Chromosomes also behave in a similar manner during metaphase I and II of meiosis. Explain how metaphase I and II of meiosis can result in genetic variation.

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..... [2]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)** as indicated in the question.

4 **(a)** Outline the conversion of light energy to chemical energy by photosynthesis. [8]

(b) Discuss the ethical implications of stem cell research and how induced pluripotent stem cells (iPSCs) overcome some of the issues. [7]

[Total: 15]

5 **(a)** Describe the multistep development of cancer. [8]

(b) Explain the significance of membrane fluidity in biological processes. [7]

[Total: 15]





